Course Name: Microprocessor and Microcontroller
[Credits: 3; Lecture Hours: 45]

Unit-I [11 Lecture Hours]
Introduction to Microprocessors: History of Microprocessors, Applications, Speed, Word Size, Memory Capacity, Classification of Microprocessors (Mention of Different Microprocessors being used), Microcomputer, Basic Block Diagram of Microcomputer, CPU, ALU, Control Unit, Buses, Memory, Input/Output.
8085 Microprocessor: Features, Architecture, Functional Block Diagram, General Purpose Registers, Register Pairs, Flags, Special Purpose Resisters, Stack Pointer, Program Counter, Types of Buses, Multiplexed Address Bus and Data Bus, Generation of Control Signals, Pin-out Diagram and Pin Description, Basic Memory and I/O Interfacing Concepts, Memory Mapped I/O and I/O Mapped I/O, Partial and Full Address Decoding.

Unit-II [14 Lecture Hours]
8085 Instructions and Programming: Operation Codes, Operands and Mnemonics, Instruction Set, Addressing Modes, Instruction Format, Instruction Classification, Data Transfer Instructions, Arithmetic Instructions, Increment and Decrement Instructions, Logical Instructions, Branch Instructions and Machine Control Instructions, Stack Operations, PUSH and PULL Instructions, Subroutine, Call and Return Instructions, Program Status Word (PSW), Delay Loops, Use of Counters, Instruction Cycle, Machine Cycle, Timing Diagrams, T-States, Op-Code Fetch, Memory Read, Memory Write, I/O Read and I/O Write, Examples of Timing Diagrams for Selective Instructions MVI and STA,Assembly Language Programming and Examples.
8085 Interrupts and Interfacing: Interrupt Structure, EI and DI Instructions, INTR and $\overline{\text{INTA}}$ Signals, Handling Multiple Interrupts using Priority Encoder, RST Instructions, Maskable and Non-Maskable Interrupts, Vectored Interrupts (TRAP, RST 7.5, 6.5 and 5.5) Non-Vectored Interrupts, SIM and RIM Instructions, Latency Time and Response Time, Direct Memory Access (DMA), HOLD and HLDA Signals, Interfacing with Analog to Digital Converter (ADC) IC 0801/0808, Interfacing using General Purpose Programmable Peripheral Interface (PPI)IC 8255, Block Diagram, Pins, Control Word, Different Modes of Operation, BSR and I/O Modes, Illustration of Application of 'Mode 0'.

Unit-III [10 Lecture Hours]
Introduction to Microcontroller: Definition and Use of Microcontroller, Difference with Microprocessor,Types of Microcontrollers, Embedded Microcontrollers, Microcontroller Architectures, Internal Block Diagram, Key Features, Pin Diagram, CISC vs. RISC Architectures, Microcontroller Memory Types, Microcontroller Features, Clocking, I/O Pins, Interrupts, Timers, Peripherals.
8051 Programming: 8051 Addressing Modes and Accessing Memory Locations using Various Addressing Modes, Assembly Language Instructions using Addressing Mode, Arithmetic and Logic Instructions, 8051 Assembly Language Programming Examples, I/O Port Programming in 8051 (using Assembly Language).

Unit-IV [10 Lecture Hours]
Introduction to Arduino Uno: Functional Block Diagram of Arduino Uno Board, ATmega328P: Microcontroller inside Arduino Uno, Functions of different Pins of Arduino Uno, Other Arduino Development Boards, Arduino IDE, Serial Monitor , Arduino Sketch, Variables, setup Function, I/O Functions: pinMode(), digitalRead(), digitalWrite(), analogRead(), analogWrite(), delay() Function, Looping Techniques, Decision Making Techniques, Designing of 1st Sketch, Arduino In-System-Programmer (ISP), Serial Port Communication with PC, Basic Interfacing and I/O Concept, Interfacing LED and Switch, Interfacing with Temperature Sensor LM35.

ELT-H-CC-3-3-P
Course Name: Microprocessor and Microcontroller Lab

[Credit: 1; Lecture Hours: 30]

Section-A: 8085 Microprocessor Assembly Language Programming

1. Program to Transfer a Block of Data.
2. Program for Addition and Subtraction of Numbers using Direct Addressing Mode.
1. Program for Addition and Subtraction of Numbers using Indirect Addressing Mode.
3. Program to Multiply Two 8-Bit Numbersusing Repeated Addition Algorithm.
4. Program to Divide Two 8-Bit Numbersusing Repeated Subtraction Algorithm.
5. Program to Divide a 16-Bit Number by 8-Bit Number.
6. Program to Search a given Number in a given List.
7. Program to Find Number of Zeroes and Ones in an 8-Bit Number.
8. Program to Find Whether an 8-Bit Number is Even or Odd.
9. Use of CALL and RETURN Instruction.
10. Program to Implement a Subroutine Operation.
11. Program to Implement a Stack Operation.
12. Program to Generate Fibonacci Series.
13. Program to Find Minimum and Maximum among N Numbers.
14. Program to Find Square Root of an Integer.
15. Program to Find GCD of Two Numbers.
16. Program to Sort Numbers in Ascending/Descending Order.
17. Program to Verify the Truth Table of Logic Gates.

Section-B: 8051 Microcontroller Assembly Language and Arduino Programming

1. Program to Find Whether the Numbers are Prime or Not.
2. Program to Find Factorial of a Number.
3. Program to Find (a) Largest of N Numbers and (b) Smallest of N Numbers.
4. Program to Arrange Numbers in Ascending/Descending Order.
5. Write and Execute a Sketch in Arduino Uno IDE to Blink an LED.
6. Write and Execute a Sketch in Arduino Uno IDE to Interface a Switch to Turn an LED On and Off.
7. Write and Execute a Sketch in Arduino Uno IDE to Interface with Ttemperature Controller IC LM35 to Monitor Temperature.

Reference Books:

- Gaonkar, Microprocessor Architecture, Programming and Applications with the 8085, Penram.
- B. Ram, Fundamentals of Microprocessors and Microcomputers, Dhanpat Rai.
- Mazidi, McKinlay, Mazidi and Das, Microprocessors and Microcontrollers, Pearson.
- Mathur and Panda, Microprocessors and Microcontrollers, PHI.
- Krishna Kant, Microprocessors and Microcontrollers: Architecture, Programming and System Design, PHI.
- Kumar, Saravanan, and Jeevananthan, Microprocessors and Microcontrollers, Oxford.
- Shah, 8051 Microcontrollers: MCS 51 Family and its Variants, Oxford.
- Ashwin Pajankar,ARDUINO Made Simple, BPB Publications.

CC-4: Mathematical Foundation, Numerical Analysis and Scilab
[Credits: 4 (3TH+1P)]

ELT-H-CC-4-3-TH
Course Name: Mathematical Foundation, Numerical Analysis and Scilab
[Credits: 3; Lecture Hours: 45]

Course Name: Electronic Communication
[Credits: 3; Lecture Hours: 45]

Unit-I [12Lecture Hours]

Introduction to Electronic Communication: Means and Modes, Block Diagram of ElectronicCommunication System, Electromagnetic Spectrum, IEEE Band Designations and Applications,Brief Idea of Frequency Allocation for Radio Communication System in India (TRAI),Need forModulation, Concept of Channels and Base-Band Signals, Concept of Bandwidth, Concept of Transmission Modes (Simplex, Half Duplex, Full Duplex).

Noise in Communication Systems: Concept of Noise, Types of Noise,Internal and External Noises, Thermal and White Noise, Signal-to-Noise (S/N) Ratio and Noise Figure, Figure of Merit,Noise Temperature.

Amplitude Modulation: Definition, Mathematical Representation of AM,Time Domain and Frequency Domain (Frequency Spectrum) Representations,Modulation Index and PercentageModulation,Concept of Under, Over and Critical Modulation, Frequency Spectrum, SidebandFrequencies, Bandwidth Requirements, Power Carried by Carrier and Sidebands, Concept of Multitone Modulating Signals, Generation of AM (Linear and Non Linear Methods), Amplitude Demodulation (Envelope Detector), Double Side Band Suppressed Carrier (DSBSC), Chopper (Ring) and Balanced Modulators, Single Side Band Suppressed Carrier (SSBSC), Phase Shift (Discrimination) Method, Pilot Carrier Amplitude Modulation, Concepts of Vestigial Side Band (VSB) Modulation and Quadrature Amplitude Modulation (QAM), Independent Side Band Modulation, Block Diagram of AM Transmitter and Receiver, Utility of Heterodyning, Super Heterodyning Principle,Super Heterodyne Receiver (SHR), Advantages over Tuned Radio Frequency (TRF) Receiver, Different Blocks of SHR, Intermediate Frequency, Local Oscillator Frequency, Image Frequency.

Unit-II [10Lecture Hours]

Frequency and Phase Modulation: Definition, Mathematical Representation of FM andPM,Equivalence between FM and PM, Time Domain and Frequency Domain (Frequency Spectrum) Representations, Modulation Index,Bandwidth Requirements, Frequency Deviation, Narrow Band FM (NBFM) and Wide Band FM (WBFM), Carson's Rule, Direct (Varactor Diode) and Indirect (Armstrong) Methods of FM Generation, FM Detection using Slope Detector and PLL Detector,Block Diagram of FM Transmitter and Receiver, Comparison between AM, FM and PM.

Analog Pulse Modulation: Channel Capacity, Sampling Theorem, Sampling and Reconstruction of Signals,Pulse Amplitude Modulation (PAM), Pulse Width Modulation (PWM), Pulse Position Modulation (PPM), Generation and Detection Techniques, Time Division Multiplexing (TDM) and Frequency Division Multiplexing (FDM).

Unit-III [11 Lecture Hours]

Digital Modulation Techniques: Need for Digital Transmission,Block Diagram of Digital Transmission and Reception,Characteristics of Data Transmission Circuits,Advantages and Disadvantages of Digital Communication, Information Capacity,Shannon Limit for Information Capacity, Bandwidth Requirements, Pulse Code Modulation (PCM), Quantizing, Uniform and Non-Uniform Quantization, Quantization Noise, Companding, Coding, Decoding, Regeneration, Data Transmission Speed (Bit Rate and Baud Rate), Noise, Cross Talk, M-Array Coding, Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), Phase Shift Keying (PSK), Binary Phase Shift Keying (BPSK) and Quadrature Phase Shift Keying (QPSK).

Unit-IV [12 Lecture Hours]

Radio Wave Propgation: Characteristics of Electromagnetic Wave, Different Modes of Radio Wave Propagation, Ground (Surface) Wave Propagation, Space (Tropospheric) Wave Propagation, Sky (Ionospheric) Wave Propagation, Structure of Atmosphere, Critical Frequency,

Maximum Usable Frequency (MUF), Virtual Height, and Skip Distance, Duct Propagation (Qualitative Discussions Only).

Cellular Communication: Concept of Cellular Mobile Communication, Frequency Bands used in Cellular Communication, Concept of Cell Sectoring and Cell Splitting, Absolute RF Channel Numbers (ARFCN), Frequency Reuse, Roaming and Hand Off, Authentication of SIM Card of Subscribers, IMEI Number, Need for Data Encryption, Architecture (Block Diagram) of Cellular Mobile Communication Network, Concept of GSM, CDMA, TDMA and FDMA, Comparison of TDMA and FDMA Technology, Simplified Block Diagram of Cellular Phone Handset, Comparative Study of GSM and CDMA, Qualitative concepts of 2G, 3G, 4G and 5G, Qualitative Idea of GPS Navigation System.

ELT-H-CC-5-4-P
Course Name: Electronic Communication Lab
[Credit: 1; Lecture Hours: 30]

Implementation with Hardware and Circuit Simulation Software

1. Study of Amplitude Modulation.
2. Study of Amplitude Demodulation.
3. Study of Frequency Modulation.
4. Study of Frequency Demodulation.
5. Study of Pulse Amplitude Modulation.
6. Study of Pulse Width Modulation.
7. Study of Pulse Position Modulation.
8. Study of Pulse Code Modulation.
9. Study of Amplitude Shift Keying.
10. Study of Phase Shift Keying.
11. Study of Frequency Shift Keying.

Reference Books:

- Kennedy, Electronic Communication Systems, Tata McGraw Hill.
- Frenzel, Principles of Electronic Communication Systems, Tata McGraw Hill.
- Tomasi, Advanced Electronic Communications Systems, Pearson.
- Roddy and Coolen, Electronic Communications, Pearson.
- Haykin, Communication Systems, Wiley.
- Lathi and Ding, Modern Digital and Analog Communication Systems, Oxford.
- Couch, Digital and Analog Communication Systems, Pearson.
- Kundu, Analog and Digital Communications, Pearson.
- Blake, Electronic Communication Systems, Cengage.
- T. L. Singal, Analog and Digital Communication, Tata McGraw Hill.

CC-6: Signals and Systems
[Credits: 4 (3TH+1P)]

ELT-H-CC-6-4-TH